Jiahao Xu

Curriculum Vitae

PhD Student, Computer Science | Virginia Tech
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Research Interests

Visual analytics, human–AI collaboration, and spatial sensemaking. I develop interactive systems that help people verify and revise AI-generated information, and investigate how AI agents can interpret and act within shared 2D workspaces.

Education

Virginia Tech — Aug. 2023 – present
PhD in Computer Science & Applications, in progress — Blacksburg, VA
Advisor: Chris North

Tufts University — Jan. 2021 – May 2023
MS in Computer Science — Medford, MA
Advisor: Remco Chang

Chang'an University — Sept. 2016 – June 2020
BE in Computer Science and Technology — Xi'an, China
Exchange study, University of California, Irvine — Apr. – Oct. 2019

Publications

Peer-reviewed publications

[1] CanvasObs: Structured Spatial Observation for Agent Editing on 2D Infinite Canvases.
Jiahao Xu, Haohui Wang, Rebecca Faust, Dawei Zhou, and Chris North.
VISxGenAI, 2026. Workshop at IEEE VIS.

[2] Spatial Visual Analytics for Multi-Document Summary Verification.
Jiahao Xu, Wei Liu, Yang Liu, Eric Krokos, Kirsten Whitley, Xuan Wang, Rebecca Faust, and Chris North.
IEEE VIS, 2026. Full paper.

[3] Drag, Infer, Reproject: Grounding LLMs through Spatial Interaction for Image Clustering.
Yang Liu, Xuxin Tang, **Jiahao Xu**, and Chris North.
IEEE VIS, 2026. Short paper.

[4] The Effects of Belief Elicitation in Visual Data Analysis: A Longitudinal Classroom Study.
Edward W. He, Vanessa Bellotti, Alexandra Scott, **Jiahao Xu**, Ashley Suh, Jen Rogers, and Remco Chang.
IEEE Transactions on Visualization and Computer Graphics, 32(2):2379–2393, 2026.

Preprints

[5] Overcoming Prior Barriers: Supervised Fine-Tuning under Long-Tail Distribution.
Haohui Wang, **Jiahao Xu**, Wangzhi Zhan, Tong Zeng, Dongqi Fu, Hong Li, Swastik Roy, Naren Ramakrishnan, Chris North, Jian Kang, Yujun Yan, and Dawei Zhou.
arXiv preprint, 2026.

Teaching Experience

Virginia Tech — Teaching Assistant
CS 5764: Information Visualization
CS 3724: Human-Computer Interaction

Tufts University — Teaching Assistant
CS 178: Visual Analytics
CS 170: Computational Theory
CS 114: Network Security